

SIMATS ENGINEERING

COURSE NAME: FUNDAMENTALS OF DATA SCIENCE

COURSE CODE: DSA0404

CO 3 - ASSESSMENT TOOL 2: Data Interpretation

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1. Continuous Distribution — Area Interpretation

A continuous random variable X is normally distributed with:

Mean, $\mu = 100$

Standard deviation, $\sigma = 15$

(a) Find the proportion of values between 85 and 130

We first convert both values into z-scores.

Formula:

$$z = (X - \mu) / \sigma$$

For X = 85:

$$z_1 = (85 - 100) / 15$$

$$z_1 = -15 / 15$$

$$z_1 = -1$$

For X = 130:

$$z_2 = (130 - 100) / 15$$

$$z_2 = 30 / 15$$

$$z_2 = 2$$

Therefore, we need the area between $z = -1$ and $z = 2$.

From the standard normal table:

Area to the left of $z = 2 = 0.9772$

Area to the left of $z = -1 = 0.1587$

Therefore,

$$P(85 < X < 130) = 0.9772 - 0.1587$$

$$P(85 < X < 130) = 0.8185$$

Converting to percentage:

$$0.8185 \times 100 = \mathbf{81.85\%}$$

Answer:

Proportion of values between 85 and 130 = 0.8185

Percentage = 81.85%

Therefore, approximately **81.85% of the values lie between 85 and 130.**

(b) A value of 145 is observed

We calculate how many standard deviations 145 is away from the mean.

Formula:

$$z = (X - \mu) / \sigma$$

Substituting:

$$z = (145 - 100) / 15$$

$$z = 45 / 15$$

$$\mathbf{z = 3}$$

Therefore, 145 is **3 standard deviations above the mean.**

Empirical Rule

According to the empirical rule for a normal distribution:

- Approximately 68% of values lie within ± 1 standard deviation.
- Approximately 95% of values lie within ± 2 standard deviations.
- Approximately 99.7% of values lie within ± 3 standard deviations.
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Since 145 is exactly **3 standard deviations above the mean**, it lies at the extreme boundary of the usual range.

It would therefore be reasonable to **flag it as a potential outlier for further investigation**, although being at 3 standard deviations does not automatically prove that it is an outlier.

Final Answer:

$$z = 3$$

The value 145 is **3 standard deviations above the mean** and should be **flagged as a potential outlier** under the empirical-rule approach.

2. Sampling Bias vs Sampling Error

Given:

Sample mean = ₹18,000

Known population mean = ₹14,500

Difference:

$$₹18,000 - ₹14,500 = ₹3,500$$

The survey samples only **online panel respondents**.

(a) Is this sampling error or sampling bias?

The discrepancy is better explained by **sampling bias**. The reason is that the survey does not give every household an equal chance of being selected. Only people who are part of an online panel are included. Some households may not participate in online panels, and these households may have different expenditure patterns.

Therefore, the sample may not properly represent the entire population.

This creates a **systematic difference** between the sample and the population.

The sample mean is:

₹18,000

while the population mean is:

₹14,500

The difference is: ₹3,500

This difference is likely caused by the **biased sampling method** rather than random variation alone.

Answer:

The discrepancy is better explained by sampling bias.

(b) Would increasing the sample size fix the problem?

No.

Increasing the sample size does not necessarily remove sampling bias.

For example, suppose the survey increases the sample from 100 online respondents to 10,000 online respondents. Even though the sample becomes much larger, all respondents are still being selected from the same online panel.

Therefore, the sample may still fail to represent households that are not part of the online panel. Increasing sample size mainly reduces **sampling error**, which is the random variation that occurs when a sample is selected. It does not automatically eliminate **sampling bias**.

Final Answer:

- Increasing the sample size would **not fix the main problem** because the problem is the biased sampling method.
- To reduce the bias, the company should use a **representative random sampling method** that gives different types of households an appropriate chance of being selected.

3. Covariance Sign vs Correlation Strength

Given:

Pair 1:

$$\text{Cov}(X,Y) = 850$$

Pair 2:

$$\text{Cov}(X,Y) = 12$$

Both covariance values are positive.

Step 1: Understand what the sign means

A positive covariance indicates that the two variables tend to move in the **same direction**.

Therefore:

Pair 1 → Positive relationship

Pair 2 → Positive relationship

However, we cannot say that Pair 1 has a stronger relationship simply because:

$$850 > 12$$

Why?

- ✓ The magnitude of covariance depends on the **units and scale of the variables**.
- ✓ For example, if one variable is measured in rupees and another in kilograms, the covariance can have a much larger numerical value than when the variables are measured in smaller units.
- ✓ Therefore, covariance values from different variable pairs cannot be directly compared to determine the strength of relationships.

What information is missing?

We need the **standard deviations of the variables**.

Using the standard deviations, covariance can be standardized into **Pearson's correlation coefficient**.

Formula:

$$r = \text{Cov}(X,Y) / (\sigma_X \times \sigma_Y)$$

Correlation always lies between:

$$-1 \leq r \leq +1$$

The absolute value of r tells us the strength of the relationship, while its sign tells us the direction.

Example:

- Suppose Pair 1 has a large covariance of 850 but also has very large standard deviations.
- Pair 2 may have covariance of only 12 but much smaller standard deviations.
- After standardization, Pair 2 could actually have a stronger correlation than Pair 1.

Final Answer:

We cannot conclude that Pair 1 has a stronger relationship because covariance depends on the **scale and units of measurement**. The missing information is the **standard deviations of both variables**, which are needed to calculate Pearson's correlation coefficient.

4. Estimation — Confidence Interval Reasoning

Given:

Sample size, $n = 100$

Sample mean = 250

Sample standard deviation = 40

Confidence level = 95%

Step 1: Calculate the Standard Error

Formula:

$$\text{Standard Error} = s / \sqrt{n}$$

Substituting:

$$\text{Standard Error} = 40 / \sqrt{100}$$

$$\text{Standard Error} = 40 / 10$$

$$\text{Standard Error} = 4$$

Step 2: Find the Critical Value

For a 95% confidence interval, the commonly used z-value is:

$$z = 1.96$$

Step 3: Calculate the Margin of Error

Formula:

$$\text{Margin of Error} = z \times \text{Standard Error}$$

$$\text{Margin of Error} = 1.96 \times 4$$

$$\text{Margin of Error} = 7.84$$

Step 4: Calculate the Confidence Interval

Formula:

$$\text{Confidence Interval} = \text{Sample Mean} \pm \text{Margin of Error}$$

Lower limit:

$$250 - 7.84 = 242.16$$

Upper limit:

$$250 + 7.84 = 257.84$$

Therefore,

$$\text{95\% Confidence Interval} = (242.16, 257.84)$$

Final Answer:

$$\text{95\% Confidence Interval} = 242.16 \text{ to } 257.84$$

Interpretation:

- The interval gives a range of values that is consistent with the sample data for the population mean.
- In repeated sampling, if we constructed confidence intervals in the same way many times, approximately **95% of those intervals would contain the true population mean**.
- It does **not** mean that there is a 95% probability that the particular population mean is between 242.16 and 257.84.
- The population mean is a fixed value. The confidence interval is the range estimated from the sample.

Final Answer:

The estimated population mean lies in the interval:

$$242.16 \leq \mu \leq 257.84$$

with a **95% confidence level**.

5. Integrated Case Study

Daily sales data in ₹'000 for 15 days:

120, 125, 118, 130, 122, 128, 500, 119, 124, 121, 126, 123, 117, 129, 125

The value **500** appears to be much larger than the other observations.

(a) Identify the Outlier Using IQR and Z-Score Methods

Step 1: Arrange the Data in Ascending Order

117, 118, 119, 120, 121, 122, 123, 124, 125, 125, 126, 128, 129, 130, 500

There are 15 observations.

Step 2: Calculate Q1 and Q3

Lower half:

117, 118, 119, 120, 121, 122, 123

Median of lower half:

$$\mathbf{Q1 = 120}$$

Upper half:

125, 126, 128, 129, 130, 500

Using the standard percentile method:

$$\mathbf{Q3 = 127}$$

Therefore,

$$\mathbf{IQR = Q3 - Q1}$$

$$\text{IQR} = 127 - 120$$

$$\mathbf{IQR = 7}$$

Step 3: Calculate the Lower and Upper Fences

Formula:

$$\mathbf{\text{Lower Fence} = Q1 - 1.5 \times \text{IQR}}$$

$$\text{Lower Fence} = 120 - (1.5 \times 7)$$

$$\text{Lower Fence} = 120 - 10.5$$

$$\mathbf{\text{Lower Fence} = 109.5}$$

Formula:

$$\mathbf{\text{Upper Fence} = Q3 + 1.5 \times \text{IQR}}$$

$$\text{Upper Fence} = 127 + (1.5 \times 7)$$

$$\text{Upper Fence} = 127 + 10.5$$

$$\text{Upper Fence} = 137.5$$

Therefore, any value below 109.5 or above 137.5 is considered an outlier.

The value 500 is greater than 137.5.

Therefore:

500 is an outlier according to the IQR method.

Step 4: Use the Z-Score Method

First calculate the mean including the outlier.

Sum of all values:

$$\begin{aligned} &120 + 125 + 118 + 130 + 122 + 128 + 500 + 119 + 124 + 121 + 126 + 123 + 117 + 129 + 125 \\ &= 2227 \end{aligned}$$

Number of observations:

$$n = 15$$

Mean:

$$\text{Mean} = 2227 / 15$$

$$\text{Mean} \approx 148.47$$

The standard deviation is approximately:

$$\sigma \approx 94.55$$

Now calculate the z-score of 500.

Formula:

$$z = (\text{X} - \text{Mean}) / \text{Standard Deviation}$$

$$z = (500 - 148.47) / 94.55$$

$$z = 351.53 / 94.55$$

$$z \approx 3.72$$

Therefore, the value 500 is approximately **3.72 standard deviations above the mean.**

A common rule is that values with:

$$|z| > 3$$

can be considered potential outliers.

Since:

$$3.72 > 3$$

500 is an outlier according to the z-score method as well.

Final Answer:

IQR method:

$$\text{Upper Fence} = 137.5$$

$$500 > 137.5$$

Therefore, **500 is an outlier.**

Z-score method:

$$z \approx 3.72$$

Since $|z| > 3$, **500 is an outlier.**

Both methods identify **500** as a clear outlier.

(b) Calculate Mean and Median With and Without the Outlier

With the Outlier

Data contains 15 observations.

$$\text{Sum} = 2227$$

Mean

Formula:

$$\text{Mean} = \Sigma X / N$$

$$\text{Mean} = 2227 / 15$$

$$\text{Mean} \approx 148.47$$

Median

The sorted data is:

117, 118, 119, 120, 121, 122, 123, 124, 125, 125, 126, 128, 129, 130, 500

There are 15 observations.

Median position:

$$(15 + 1) / 2 = 8\text{th observation}$$

The 8th observation is 124.

Therefore,

$$\text{Median} = 124$$

With Outlier:

$$\text{Mean} \approx 148.47$$

$$\text{Median} = 124$$

Without the Outlier

Remove 500.

Remaining data:

117, 118, 119, 120, 121, 122, 123, 124, 125, 125, 126, 128, 129, 130

Number of observations:

$$N = 14$$

New sum:

$$2227 - 500 = 1727$$

Mean

$$\text{Mean} = 1727 / 14$$

Mean ≈ 123.36

Median

There are 14 observations.

Median is the average of the 7th and 8th observations.

7th observation = 123

8th observation = 124

Therefore,

Median = $(123 + 124) / 2$

Median = 123.5

Without Outlier:

Mean ≈ 123.36

Median = 123.5

Comparison

Measure	With Outlier	Without Outlier
Mean	148.47	123.36
Median	124	123.5

The outlier causes the mean to increase from approximately **123.36 to 148.47**.

However, the median changes only slightly from **123.5 to 124**.

Interpretation:

- The **mean is highly sensitive to outliers** because it uses every value in its calculation.
- The **median is much less sensitive to outliers** because it depends mainly on the middle position of the data.
- Therefore, the median gives a more representative measure of the typical daily sales when extreme values are present.

(c) Calculate Skewness With the Outlier Included

The data contains one extremely large value, 500, compared with the remaining observations, which are mostly between 117 and 130.

We use the standardized third-moment formula for skewness:

$$\text{Skewness} = \Sigma(X - \bar{X})^3 / [N \times \sigma^3]$$

For the given data:

Mean ≈ 148.47

Standard deviation ≈ 94.55

After calculating the deviations and applying the formula:

Skewness ≈ 3.46

Interpretation:

The skewness value is:

$3.46 > 0$

Therefore, the distribution is **positively skewed (right-skewed)**. The very large value of 500 creates a long tail toward the right side of the distribution. This is also consistent with the fact that:

$$\text{Mean} \approx 148.47 > \text{Median} = 124$$

In a positively skewed distribution, the mean is generally pulled toward the larger values.

Final Answer:

Skewness ≈ 3.46

Distribution = Positively skewed / Right-skewed

The high sales value of 500 is responsible for the strong right skew.

(d) Recommendation: Remove, Cap, or Investigate the Outlier?

The value 500 should **not be automatically removed**.

Instead, the company should **investigate the outlier first**.

Statistical Evidence:

1. The IQR method identifies 500 as an outlier because:

$$500 > \text{Upper Fence of } 137.5$$

2. The z-score method also identifies it as an outlier:

$$z \approx 3.72$$

Since the absolute z-score is greater than 3, it is an extreme observation.

3. The mean is strongly affected:

With outlier:

$$\text{Mean} \approx 148.47$$

Without outlier:

$$\text{Mean} \approx 123.36$$

The difference is approximately:

$$148.47 - 123.36 = 25.11$$

4. The median is barely affected:

With outlier:

$$\text{Median} = 124$$

Without outlier:

$$\text{Median} = 123.5$$

Therefore, the median changes by only:

$$124 - 123.5 = 0.5$$

5. The skewness is very high:

$$\text{Skewness} \approx 3.46$$

This indicates strong positive/right skewness caused by the extreme value.

Recommendation:

The company should **investigate the value 500 before deciding what to do with it.**

If 500 represents a genuine event, such as a major bulk order, festival sale, special promotion, or unusually successful business day, it should **not be removed**. It is a real observation and may be important for understanding business performance.

If 500 is caused by a data-entry error or measurement error, it should be corrected or removed.

Capping should generally not be the first choice because it changes a genuine observation into an artificial value.

Final Recommendation:

- **Investigate first → Verify the cause → Then decide whether to retain or correct it.**
If 500 is a genuine sale:
- **Retain the value and report the median or provide additional outlier-aware statistics.**
If 500 is a data error:
- **Correct or remove the erroneous value.**

The statistical evidence strongly supports investigating the value because both the **IQR and z-score methods identify it as an outlier**, while the **mean and skewness are substantially distorted by it.**